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Course of Study Industrial Artificial Engineering

Topic

Bearing Remaining Useful Life Prediction Using Robust Multi-Branch Deep Learning

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Abstract

This report summarizes a bearing remaining useful life (RUL) prediction project completed using the Cross-Industry Standard Process for Data Mining (CRISP-DM). The project uses the PRONOSTIA bearing degradation dataset, which records run-to-failure vibration data under multiple operating conditions, and it follows the Robust Multi-Branch Deep Learning (Robust-MBDL) idea proposed by Tran, Vu, Pham, Boudaoud, and Nguyen (2024). The pipeline combines data understanding, preprocessing, feature extraction, model design, evaluation, and deployment planning. The adopted model uses three complementary streams: hand-crafted one-dimensional vibration features, two-dimensional time-frequency spectrograms, and denoised raw vibration signals. An LSTM autoencoder is used for noise filtering before feature extraction, and attention-augmented bidirectional LSTM blocks are used to fuse information for RUL regression and operating-condition classification. This report also includes phase-wise questions and answers explaining what was done, why the approach was selected, and how the method supports predictive maintenance. Finally, the team work distribution is mapped to four project blocks: environment/data preparation, data loading, visualization/explainability, and modeling/evaluation.

Keywords: remaining useful life, predictive maintenance, PRONOSTIA, CRISP-DM, LSTM autoencoder, multi-branch deep learning

Introduction and Phase 1: Business Understanding

Rolling-element bearings are critical components in rotating machinery. Unexpected bearing failure can stop production, increase repair cost, create safety risk, and reduce machine availability. Remaining useful life prediction addresses this problem by estimating how long a component can continue operating before failure. In predictive maintenance, RUL is useful because it helps maintenance teams replace parts before catastrophic failure while avoiding unnecessary early replacement. The Robust-MBDL research paper positions accurate RUL estimation as a central task for predictive maintenance of rotating machines and proposes a deep learning framework that uses multiple views of vibration data (Tran et al., 2024).

The business objective of this project is to build a data-driven RUL prediction workflow for bearings using vibration sensor data. The data-mining objective is to learn degradation patterns from PRONOSTIA run-to-failure trials and predict normalized RUL values between 0 and 1. A secondary objective is to classify the operating condition or degradation state so that predictions can be interpreted as maintenance actions. This goal fits CRISP-DM because the project starts from a practical maintenance problem, translates it into data-mining targets, builds and evaluates a model, and ends with deployment recommendations (Chapman et al., 2000).

Objective	Project meaning	Why it matters
RUL regression	Predict the remaining lifetime trajectory from vibration windows.	Provides early warning and supports maintenance scheduling.
Operating-condition awareness	Account for different load and speed conditions in the bearing trials.	Improves generalization across multiple machine settings.
Robustness to noise	Use denoising and multi-stream inputs rather than one raw signal only.	Industrial sensors are noisy and fault signatures can be weak.
Actionable reporting	Convert numerical predictions into status classes and visual outputs.	Makes model results easier for engineers and managers to use.

Phase 2: Data Understanding

The project uses PRONOSTIA, an experimental platform for accelerated bearing degradation tests. The dataset was introduced as a benchmark for prognostics and health management and contains run-to-failure vibration data for bearings under controlled operating conditions (Nectoux et al., 2012). In the project implementation, the data loader expects a Learning_set and Test_set structure with folders such as Bearing1_1 and Bearing1_3. Each vibration file is treated as one recording window, and elapsed time is computed from the window index.

Data element	Value used in project	Purpose
Sensors	Horizontal and vertical accelerometers	Capture vibration changes caused by bearing degradation.
Sampling rate	25,600 Hz	High-frequency sampling preserves fault-related vibration content.
Window size	2,560 samples per file	Creates short signal windows for feature extraction and model input.
Recording interval	One 0.1-second recording every 10 seconds	Represents degradation over time while limiting data volume.
Operating conditions	Three combinations of load and speed	Tests whether the model can generalize across machine regimes.
Failure criterion	Accelerometer signal above 20g	Defines end of life for RUL label construction.

Exploratory analysis focuses on lifetime spread, vibration energy, and statistical indicators. Root mean square (RMS) is useful because increasing vibration energy often accompanies degradation. Kurtosis is useful because bearing defects can produce impulsive peaks before end-of-life. These data-understanding activities justify the later use of time-domain, frequency-domain, and time-frequency features rather than relying only on raw samples.

Phase 3: Data Preparation

Data preparation is the core engineering phase. The Robust-MBDL paper defines the overall workflow as noise filtering, feature extraction, health-indicator construction, and multi-branch learning (Tran et al., 2024). The project follows the same logic. First, data are loaded and grouped by bearing. Second, raw vibration signals are denoised with an LSTM autoencoder. Third, three feature streams are built: 14 scalar one-dimensional features, continuous wavelet transform spectrograms, and fixed-length denoised raw signals.

The LSTM autoencoder is selected because LSTM networks are designed to learn time dependencies in sequential data, and autoencoders can learn compact representations by reconstructing their inputs. For vibration signals, this makes the autoencoder a practical unsupervised noise filter before RUL modeling (Hochreiter & Schmidhuber, 1997; Tran et al., 2024).

Feature stream	What was prepared	Why this approach was used
1D statistical and spectral features	RMS, variance, kurtosis, skewness, crest factor, peak value, peak-to-peak, shape factor, impulse factor, clearance factor, margin factor, FFT peak-to-peak, FFT energy, and PSD.	Compact features summarize vibration energy, spread, impulsiveness, and spectral power. They are easy to compute and interpretable.
2D CWT spectrograms	Morlet-wavelet time-frequency maps resized to model input size.	Fault signals are non-stationary; CWT captures how frequency content changes within a window.
Denoised raw signal	Autoencoder output truncated or padded to a fixed length.	Keeps transient information that may be lost in scalar feature aggregation.
RUL labels	HI-FPT and HI-PCA label construction.	FPT provides a degradation start point; PCA health indicators provide smoother learning targets.

For label construction, HI-FPT uses the first prediction time concept to keep RUL constant during the healthy region and then decay toward failure. HI-PCA uses principal-component movement across time to create a smoother degradation index. The smoother HI-PCA trajectory is preferred for stable regression training because it provides gradual changes instead of sharp label transitions.

Phase 4: Modeling

The model is based on a multi-branch deep learning architecture. Tran et al. (2024) argue that vibration data should be represented in multiple forms because raw data, statistical features, and time-frequency representations capture different parts of degradation behavior. Their Robust-MBDL architecture uses three branches and fuses them through attention-based bidirectional LSTM components. This project adopts the same design idea and implements the model in PyTorch.

Branch	Input	Model component	Reason
Branch 1	Fourteen 1D features	1D convolutional layers followed by AB-LSTM	Learns local relationships among engineered features and temporal degradation context.
Branch 2	CWT spectrogram image	ResNet-style 2D convolutional blocks	Uses image-like time-frequency patterns; skip connections support deeper learning (He et al., 2016).
Branch 3	Denoised raw signal	1D convolutional building blocks	Preserves raw transient impulses and reduces information loss.
Fusion	Outputs of all branches	Attention-augmented bidirectional LSTM and final heads	Attention can emphasize the most relevant timesteps and feature representations (Vaswani et al., 2017).

The prediction heads are designed for two tasks: RUL regression and operating-condition classification. This multi-task structure is useful because operating conditions influence degradation rate. The RUL head outputs a normalized lifetime value through a sigmoid activation, and the classification head outputs condition probabilities through a softmax layer. Training uses regression metrics such as RMSE and MAE and classification accuracy for operating condition or degradation class.

Phase 5: Evaluation and Phase 6: Deployment Planning

Evaluation checks whether the model is technically accurate and practically useful. The main regression metrics are root mean squared error (RMSE), mean absolute error (MAE), and percentage error per bearing. Classification accuracy is also monitored for operating condition or degradation class. Cross-validation is used to reduce the chance that results depend on a single split, and scaler fitting is restricted to training folds to avoid data leakage.

The current project run trained for 300 epochs. The implementation report indicates that the model was still undertrained at that point, with systematic RUL overestimation and low degradation-class accuracy. The most important recommendation is to train for more epochs, use early stopping, prefer smoother HI-PCA labels, apply dynamic joint loss weighting, and add small Gaussian-noise augmentation. These changes are consistent with the Robust-MBDL paper, which emphasizes robustness, denoising, and multi-source vibration representations (Tran et al., 2024).

Deployment planning should convert the notebook into a repeatable monitoring pipeline. In production, new vibration windows would be read from sensors, denoised, transformed into the same three feature streams, scaled using the saved training scaler, and passed through the trained model. The output should be displayed as predicted RUL, confidence/status class, recent trend plot, and a maintenance recommendation. A deployed system should also log predictions, compare predicted and actual failures when available, and trigger retraining when data distribution changes.

Recommended deployment practices are to report RMSE, MAE, percentage error, condition accuracy, and class match rate; use SHAP or input-gradient analysis to identify important feature streams (Lundberg & Lee, 2017); test noise sensitivity before deployment; map RUL outputs to maintenance labels; and monitor data drift, prediction error, and confidence after deployment.

Work Distribution

The work distribution below follows the block mapping provided for the project and aligns the blocks with the code functions and CRISP-DM responsibilities shown in the attached summary image.

Member	**Primary responsibility**	**Key deliverables**
Monica Subin	Environment setup and data preparation	Imported libraries; set seeds and CPU/GPU device; defined feature lambda functions; prepared scaling, PCA, and noise-augmentation helpers.
Uday Maduri	Data understanding and loading	Configured PRONOSTIA path, sampling frequency, and sample constants; implemented load_bearing(), discover_bearings(), and the learning/test inventory.
Vinay Sankar	Modeling and evaluation	Implemented PyTorch model setup, DataLoader/TensorDataset preparation, LSTM/AB-LSTM blocks, K-fold validation, optimizer/scheduler, RMSE/MAE/accuracy, and percentage-error analysis.
Mahesh	Visualization and explainability	Created EDA plots, RMS/lifetime charts, RUL curves, predicted-vs-actual plots, feature trends, SHAP/gradient-style interpretation, and tqdm progress tracking.

This distribution gives each member a clear ownership area while keeping dependencies connected: Block 2 supplies data to Block 1 preparation functions, Blocks 1 and 2 supply inputs to Block 4 modeling, and Block 3 converts results into visual and explainable outputs for evaluation and presentation.

Conclusion

The project demonstrates an end-to-end RUL prediction workflow for bearing prognostics. Its main strength is the use of complementary vibration representations: engineered features for interpretability, CWT spectrograms for time-frequency behavior, and denoised raw signals for transient information. The Robust-MBDL-based design is appropriate because it addresses known limitations of single-input models and supports both RUL regression and operating-condition classification. The next improvements should focus on longer training, early stopping, smoother HI-PCA labels, stronger validation, and a deployment-ready preprocessing pipeline.

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